

國立東華大學應用數學系

碩士論文

時空 skew-t 過程於高門檻超標預測之延
伸：AR(2) 時間先驗與多重解析度樣條均
值

*Extending the Spatio-Temporal Skew-t Process for High-Threshold
Exceedance Prediction: AR(2) Temporal Priors and Multi-Resolution Spline
Means*



研究生：謝易軒
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摘要

在環境監測中，具政策意涵的推論目標往往不是場域的平均行為，而是在未知站點位置上其超越高門檻的機率，地面臭氧超越美國環保署 (US EPA) 75 ppb 合規門檻之預測即為典型情境。針對此問題，Morris et al. (2017) 提出一套階層式貝氏的時空 skew- t 過程 (STP)，其由四項要素構成：具 Matérn 協方差結構的 skew- t 邊際分布、用以消除遠距漸近相依的每日空間隨機分割、上尾門檻檢查，以及作用於 PIT 變換後時變潛在參數之 AR(1) 先驗。本論文以此架構為探討對象，追問在此四項要素中「何處」仍保留改善上尾預測精度的空間，並以兩個方向相反的軸線加以探測：在時間軸上，將 AR(1) 先驗推廣為符合 Yule-Walker 關係的標準化 AR(2) 先驗，以維持單位邊際變異數並保有每日 skew- t 邊際結構；在空間軸上，於迴歸均值中引入多重解析度薄板樣條 (MRTS) 基底函數，以捕捉穩態 Matérn 協方差所無法表現的非定態結構。在模擬資料與 EPA 臭氧資料上以樣本外 Brier 分數評分後，兩項延伸皆未能全面勝過 STP 基準；其價值在於將該基準擴大為一個候選模型類，可由樣本外上尾分數從中選取最適合手邊資料的設定。

關鍵字：時空 skew- t 過程、門檻超越預測、空間隨機分割、階層式貝氏推論、標準化 AR(2) 先驗、多重解析度薄板樣條、Brier 分數、臭氧



Abstract

In environmental monitoring, the policy-relevant target is often not the average behavior of a field but the probability that it exceeds a high threshold at unmonitored sites, as in the prediction of ground-level ozone exceedances of the U.S. EPA 75 ppb compliance level. For this problem Morris et al. (2017) proposed a hierarchical Bayesian spatio-temporal skew- t process (STP) built from four components: a skew- t marginal with Matérn covariance, a daily random spatial partition that removes long-range asymptotic dependence, upper-tail threshold censoring, and an AR(1) prior on the PIT-transformed time-varying latents. Taking this architecture as the object of study, we ask where among the four components room for sharper upper-tail prediction remains, and probe it along two opposing axes: on the temporal axis we generalize the AR(1) prior to a standardized, Yule–Walker–consistent AR(2) prior that preserves unit marginal variance and the daily skew- t margin; on the spatial axis we augment the regression mean with multi-resolution thin-plate spline (MRTS) basis functions—a flexible spatial trend—that capture non-stationary mean structure a constant mean would otherwise misattribute to the stationary Matérn covariance. Scored by held-out Brier scores on both simulated data and the EPA ozone record, neither extension uniformly improves on the STP baseline; rather, they enlarge it into a candidate class from which held-out tail scores can select the setting best suited to the data at hand.

Keywords: spatio-temporal skew- t process; threshold exceedance prediction; random spatial partition; hierarchical Bayesian inference; standardized AR(2) prior; multi-resolution thin-plate splines; Brier score; ozone

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1 Introduction

The need for accurate pollution-concentration estimates over broad spatial domains keeps growing. Epidemiological evidence has tied air quality to morbidity and mortality, and regulation has tightened. Direct measurement alone cannot meet this need. Its cost and logistics cap the spatial coverage and temporal resolution of any monitoring network. This gap motivates statistical methods that extend a network’s spatial reach with quantified confidence (U.S. Environmental Protection Agency, 2024). Ground-level ozone is the canonical regulatory case. A site complies only if the three-year average of its annual fourth-highest daily maximum eight-hour concentration stays at or below 75 parts per billion (ppb). The policy-relevant target is thus not the mean field but its upper tail, namely the probability that an unmonitored location breaches a high threshold. We pursue two goals: spatial prediction at unmonitored sites, and mapping the probability of extreme exceedances across the domain.

Faithfulness to the upper tail is where conventional spatial models fall short. Likelihood-based spatial prediction usually rests on a Gaussian process. A Gaussian margin is light-tailed and symmetric. It is therefore mismatched to fields that are heavy-tailed and skewed. A Gaussian process is also asymptotically independent. The probability that two sites are jointly extreme decays to zero as the level rises. A Gaussian covariance then cannot reproduce the tendency of nearby stations to breach a high ozone level on the same day. The skew- t process removes both defects at once (Morris et al., 2017). Its margin is heavy-tailed and asymmetric. It retains asymptotic dependence between sites. It also keeps computation on the order of a Gaussian model. One difficulty remains. In a spatial setting, the multivariate skew- t is asymptotically dependent between all sites, regardless of the distance between them. This imposes long-range dependence that the data do not support. Morris meets the concern with a fresh random spatial partition, drawn anew each day. The partition confines asymptotic dependence to stations near one another, and leaves distant sites effectively independent.

A separate literature models extremes directly, as exceedances over a high threshold. This is the classical route of extreme-value statistics. Univariate work fits the generalized Pareto distribution to threshold exceedances. Multivariate work adds a censored treatment of the different ways a threshold can be crossed. Spatial versions follow, fitted by censored pairwise likelihood. Later extensions reach space-time data and supply more efficient full-likelihood inference (Morris et al., 2017). All of this rests on one assumption. The threshold must be high enough for extreme-value limit theory to hold and to justify extrapolation beyond the observed range. Two costs follow. The inference is computationally intensive. In practice it is confined to rather small datasets. The ozone problem meets neither condition. The regulatory level

of interest is not extreme enough for the limit theory. The network holds 1,089 monitoring stations, well beyond the reach of these methods.

This thesis builds on the model of Morris et al. (2017) rather than replacing it. We retain its spatial skew- t process and its three extensions. These are the daily random partition, the threshold censoring, and the first-order autoregressive prior on the latent temporal parameters. We then probe the model along two axes. The first axis is temporal. The AR(1) prior ties each day only to the day before. We extend it to a second-order prior that adds one further lag. The aim is to capture longer-range temporal dependence in the record. The second axis is spatial. The baseline relies on a stationary Matérn covariance for spatial structure. Real ozone fields are not stationary in the mean. We draw on the multi-resolution thin-plate spline (MRTS) basis of Tzeng and Huang (2018). This basis is automatically ordered and depends on location alone. We incorporate it in the simplest available way, as fixed-rank covariates in the mean. This attempts to relax the stationarity assumption without altering the original model structure. Each axis is a candidate extension, not a wholesale replacement of the baseline. Together they aim to sharpen the model on threshold-exceedance prediction.

2 Methodology

In this section, we first introduce the spatio-temporal skew- t process (STP) proposed by Morris et al. (2017), and then develop two extensions of it. The first extends the autoregressive structure on the Gaussian-copula margins of the latent processes from AR(1) to AR(2). The second augments the regression component with multi-resolution thin-plate spline (MRTS) basis functions to capture non-stationary mean structure. Lastly, we present the full hierarchical model together with the MCMC algorithm used for inference.

2.1 The spatio-temporal skew- t model

Let $Y_t(\mathbf{s})$ denote the response at spatial location $\mathbf{s} \in \mathcal{D} \subset \mathcal{R}^2$ on day $t \in \{1, \dots, n_t\}$. Following Morris et al. (2017), we model

$$Y_t(\mathbf{s}) = \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s}), \quad (1)$$

where $\mathbf{X}_t(\mathbf{s})$ is a p -vector of covariates, $\boldsymbol{\beta}$ is the corresponding regression coefficient, $\lambda \in \mathcal{R}$ is a scalar skewness parameter, and $v_t(\mathbf{s})$ is a standard Gaussian process with stationary isotropic

Matérn correlation

$$\rho(h) = (1 - \gamma) I(h = 0) + \gamma \frac{1}{\Gamma(\nu) 2^{\nu-1}} \left(\sqrt{2\nu} \frac{h}{\varphi} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{h}{\varphi} \right), \quad (2)$$

where $\varphi > 0$ is the spatial range, $\nu > 0$ is the smoothness, $\gamma \in [0, 1]$ is the proportion of variance attributed to the spatial component, K_ν is the modified Bessel function of the second kind, and $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$ is the Euclidean distance between sites $\mathbf{s}_1$ and $\mathbf{s}_2$. Marginalizing over $|z_t(\mathbf{s})|$ and $\sigma_t(\mathbf{s})$, the finite-dimensional vector $\mathbf{Y}_t = [Y_t(\mathbf{s}_1), \dots, Y_t(\mathbf{s}_n)]^\top$ follows an n -dimensional skew- t distribution; consequently (1) provides asymmetric and heavy-tailed margins together with asymptotic dependence between nearby sites.

Censoring. Because the EPA compliance criterion focuses on exceedances of a high level L , the model fits only the upper tail by censoring the data at a pre-specified threshold $T \leq L$. Writing the partially censored response $\tilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, inference is based on $\tilde{\mathbf{Y}}_t = [\tilde{Y}_t(\mathbf{s}_1), \dots, \tilde{Y}_t(\mathbf{s}_n)]^\top$, and the censored values below T are imputed inside the MCMC.

Random Partition. To remove the long-range asymptotic dependence that arises when a single scalar z and σ^2 are shared across the entire domain, the model uses a daily random partition. For each day t , let $\mathbf{w}_{t1}, \dots, \mathbf{w}_{tK}$ denote K spatial knots in $\mathcal{D}$, and define the partition subregions

$$P_{tk} = \{\mathbf{s} \in \mathcal{D} : k = \arg \min_\ell \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \quad k = 1, \dots, K. \quad (3)$$

For $\mathbf{s} \in P_{tk}$ we set $z_t(\mathbf{s}) = z_{tk}$ and $\sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$, with $z_{tk} \stackrel{iid}{\sim} N(0, 1)$ and $\sigma_{tk}^2 \stackrel{iid}{\sim} \text{IG}(a/2, b/2)$. Within each subregion the model retains the skew- t form, while sites in different partitions are not asymptotically dependent.

Temporal Dependence. Daily temporal dependence is introduced on the latent parameters that drive the tail, namely the knot locations $\mathbf{w}_{tk}$, the skewness latents z_{tk} , and the scale parameters σ_{tk}^2 . To preserve the marginal skew- t distribution at every day, each parameter is first transformed to a standard normal scale via the probability integral transform (PIT):

$$w_{tki}^* = \Phi^{-1} \left[\frac{w_{tki} - \min(\mathbf{s}_i)}{\max(\mathbf{s}_i) - \min(\mathbf{s}_i)} \right], \quad i = 1, 2, \quad (4)$$

$$z_{tk}^* = \Phi^{-1} \{\text{HN}(z_{tk})\}, \quad (5)$$

$$\sigma_{tk}^{2*} = \Phi^{-1}\{\text{IG}(\sigma_{tk}^2)\}, \quad (6)$$

where Φ is the standard normal distribution function, HN is the half-normal distribution function, and IG is the inverse-gamma distribution function with parameters $(a/2, b/2)$. After transformation, each $\mathbf{w}_{tk}^*, z_{tk}^*, \sigma_{tk}^{2*}$ has a standard normal marginal distribution, and Morris et al. (2017) place a stationary AR(1) time series on each of them:

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^* \sim N_2(\phi_w \mathbf{w}_{t-1,k}^*, (1 - \phi_w^2) \mathbf{I}_2), \quad (7)$$

$$z_{tk}^* \mid z_{t-1,k}^* \sim N(\phi_z z_{t-1,k}^*, 1 - \phi_z^2), \quad (8)$$

$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*} \sim N(\phi_\sigma \sigma_{t-1,k}^{2*}, 1 - \phi_\sigma^2), \quad (9)$$

with $|\phi_w|, |\phi_z|, |\phi_\sigma| < 1$, and initial values $\mathbf{w}_{1k}^* \sim N_2(\mathbf{0}, \mathbf{I}_2)$, $z_{1k}^* \sim N(0, 1)$, $\sigma_{1k}^{2*} \sim N(0, 1)$. Transforming back through (4)–(6) gives $\mathbf{w}_{tk} \sim \text{Unif}(\mathcal{D})$, $z_{tk} \sim \text{HN}(0, 1)$, and $\sigma_{tk}^2 \sim \text{IG}(a/2, b/2)$ at every t , so that the spatial skew- t structure of (1) is preserved at each day.

2.2 Extremal dependence

For exceedance prediction, what matters is how strongly the skew- t process couples extreme values at two sites, and how that coupling changes with their separation. One measure of this extremal dependence is the χ statistic (Coles, Heffernan, and Tawn, 1999). For a stationary and isotropic spatial process, the χ statistic for two locations separated by distance h is

$$\chi(h) = \lim_{c \rightarrow c^*} \Pr[Y(\mathbf{s} + h) > c \mid Y(\mathbf{s}) > c], \quad (10)$$

where c^* is the upper limit of the support of Y ; for the skew- t distribution $c^* = \infty$. If $\chi(h) = 0$, then observations are asymptotically independent at distance h . For Gaussian processes, $\chi(h) = 0$ regardless of the distance h , so they are not suitable for modeling asymptotically dependent extremes. Unlike the Gaussian process, the skew- t process is asymptotically dependent. However, one problem with the spatial skew- t process is that $\lim_{h \rightarrow \infty} \chi(h) > 0$. This occurs because all observations, both near and far, share the same z and σ terms. Such long-range dependence is undesirable over large domains, where only local dependence is expected. Morris et al. (2017) avoid this problem through a random partitioning mechanism, described in detail in Section 2.1.

2.3 Extension to AR(2) time-varying coefficients

In this section, we generalize the AR(1) prior (7)–(9) on the transformed latents to a stationary AR(2) prior.

The AR(2) prior. Writing $\{Y_t\}$ for any scalar component of $\mathbf{w}_{tk}^*$, or for z_{tk}^* or σ_{tk}^{2*} at a fixed knot k , the prior is the stationary AR(2) recursion

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2), \quad t \geq 3, \quad (11)$$

with (ϕ_1, ϕ_2) confined to the stationarity triangle

$$|\phi_2| < 1, \quad \phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1. \quad (12)$$

Because the inverse PIT (4)–(6) returns the half-normal, inverse-gamma, and uniform target margins only when $\gamma_0 := \text{Var}(Y_t) = 1$, the innovation variance is not free: solving the Yule–Walker equations under $\gamma_0 = 1$ (Appendix A.1) fixes the standardized autocorrelations

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2}, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2, \quad (13)$$

$$\sigma_\epsilon^2 = 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2. \quad (14)$$

Setting $\phi_2 = 0$ gives $\gamma_1 = \phi_1$ and $\sigma_\epsilon^2 = 1 - \phi_1^2$, the AR(1) form of (7)–(9), so the AR(2) prior strictly nests the prior of Morris et al. (2017).

Stationary initial pair. The recursion (11) needs two seeds, and independent $N(0, 1)$ seeds would leave $\text{Var}(Y_t) \neq 1$ over the first days and break the margin. We therefore draw the initial pair from the stationary bivariate Gaussian

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix} \right), \quad (15)$$

so that $\text{Var}(Y_t) = 1$ holds from $t = 1$ onward.

Application to the three transformed parameters. Mirroring the AR(1) block (7)–(9), the AR(2) prior (11) with initial condition (15) replaces it component-wise:

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^*, \mathbf{w}_{t-2,k}^* \sim N_2 \left(\phi_1^{(w)} \mathbf{w}_{t-1,k}^* + \phi_2^{(w)} \mathbf{w}_{t-2,k}^*, \sigma_{\epsilon,w}^2 \mathbf{I}_2 \right), \quad (16)$$

$$z_{tk}^* \mid z_{t-1,k}^*, z_{t-2,k}^* \sim N \left(\phi_1^{(z)} z_{t-1,k}^* + \phi_2^{(z)} z_{t-2,k}^*, \sigma_{\epsilon,z}^2 \right), \quad (17)$$

$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*}, \sigma_{t-2,k}^{2*} \sim N \left(\phi_1^{(\sigma)} \sigma_{t-1,k}^{2*} + \phi_2^{(\sigma)} \sigma_{t-2,k}^{2*}, \sigma_{\epsilon,\sigma}^2 \right), \quad (18)$$

where each pair $(\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ lies in the stationarity triangle (12), the innovation variances follow from (13)–(14), and the initial pairs follow (15) component-wise. After the inverse PIT (4)–(6), the back-transformed $\mathbf{w}_{tk}, z_{tk}, \sigma_{tk}^2$ keep the same marginals as the AR(1) baseline, so the spatial skew- t model (1) is preserved at every t . The Gibbs-within-MH full conditional that updates each Y_t now carries three prior factors (self, lag-1, lag-2); its explicit form and the Metropolis–Hastings acceptance ratio are given in Appendix A.2.

2.4 Incorporating MRTS covariates

Tzeng and Huang (2018) proposed the multi-resolution thin-plate spline (MRTS) basis function, a basis whose functions are ordered from coarse to increasingly fine resolution. This ordering lets a model add spatial detail one resolution at a time; we define the basis below on a 2-dimensional spatial domain. Consider n distinct locations, $\mathbf{s}_1, \dots, \mathbf{s}_n$, where $\mathbf{s}_i = (x_{i1}, x_{i2})^\top \in \mathcal{R}^2$ for $i = 1, \dots, n$, and let

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{pmatrix} \quad (19)$$

. Then the MRTS basis functions $f_k(\mathbf{s})$ at a location $\mathbf{s} = (x_1, x_2)^\top$ are defined as

$$f_k(\mathbf{s}) = \begin{cases} 1, & k = 1, \\ x_{k-1}, & k = 2, 3, \\ \lambda_{k-3}^{-1} \{ \phi(\mathbf{s}) - \Phi \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x} \}^\top \mathbf{v}_{k-3}, & k = 4, \dots, n, \end{cases} \quad (20)$$

where $\mathbf{x} = (1, x_1, x_2)^\top$, $\boldsymbol{\phi}(\mathbf{s}) = (\phi_1(\mathbf{s}), \dots, \phi_n(\mathbf{s}))^\top$ with

$$\phi_i(\mathbf{s}) = \frac{1}{8\pi} \|\mathbf{s} - \mathbf{s}_i\|^2 \log(\|\mathbf{s} - \mathbf{s}_i\|), \quad i = 1, \dots, n, \quad (21)$$

and $\boldsymbol{\Phi}$ is the $n \times n$ matrix with (i, j) -th element $\phi_j(\mathbf{s}_i)$, $\mathbf{v}_k$ is the k -th column of $\mathbf{V}$, and $\mathbf{V} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{V}^\top$ is the eigen-decomposition of $\mathbf{Q} \boldsymbol{\Phi} \mathbf{Q}$ with $\lambda_1 \geq \dots \geq \lambda_n$ and $\mathbf{Q} = \mathbf{I} - \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$.

The MRTS basis enters the STP through the regression component. Fix the number of basis functions $K_{\text{MRTS}} \geq 1$ and let

$$\mathbf{f}(\mathbf{s}) = (f_1(\mathbf{s}), f_2(\mathbf{s}), \dots, f_{K_{\text{MRTS}}}(\mathbf{s}))^\top \quad (22)$$

collect the first K_{MRTS} basis functions defined in (20). Then the day- t covariate vector $\mathbf{X}_t(\mathbf{s})$ in (1) is replaced by

$$\begin{aligned} \mathbf{X}_t^*(\mathbf{s}) &= [\mathbf{f}(\mathbf{s})^\top, \mathbf{X}_t(\mathbf{s})^\top]^\top \\ &= [f_1(\mathbf{s}), f_2(\mathbf{s}), \dots, f_{K_{\text{MRTS}}}(\mathbf{s}), X_1(\mathbf{s}, t), \dots, X_p(\mathbf{s}, t)]^\top, \end{aligned} \quad (23)$$

where $X_j(\mathbf{s}, t)$ represents the j -th exogenous covariate of $\mathbf{X}_t(\mathbf{s})$. The model in (1) can then be expressed as

$$Y_t(\mathbf{s}) = \mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^* + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s}), \quad (24)$$

which is identical to the STP model (1) on all components other than the mean, with $\boldsymbol{\beta}^* \in \mathcal{R}^{K_{\text{MRTS}}+p}$. Because the MRTS basis functions depend only on location, each site $\mathbf{s}$ shares the same ordered set of basis values $\mathbf{f}(\mathbf{s})$ at every day t . The number of basis functions K_{MRTS} is the practical tuning parameter: Section 3.1 compares $K_{\text{MRTS}} \in \{3, 5, 8, 10, 12, 15, 20, 25\}$ on simulated data of different models.

2.5 Hierarchical model

Following Morris et al. (2017), we write the spatial skew- t threshold-exceedance model as a Bayesian hierarchical model, fit by Markov chain Monte Carlo (MCMC). The data are partially censored skew- t observations: at each MCMC iteration we impute the values below the threshold T , so that estimation proceeds on the completed $\mathbf{Y}$. Conditional on the partition latents $z_{tk}(\mathbf{s})$, $\sigma_{tk}^2(\mathbf{s})$ and the subregions P_{tk} , $\mathbf{Y}$ is multivariate Gaussian. The knot locations

$\mathbf{w}$ that define the partition are themselves unknown and random. Their number of knots K , by contrast, we do not also make random, since that is computationally onerous; instead we fix it at several values and select among them by held-out Brier scores (Section 3.2). The hierarchical model is

$$\begin{aligned}
Y_t(\mathbf{s}) \mid z_t(\mathbf{s}), \sigma_t^2(\mathbf{s}), P_{tk}, \Theta &= \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s}), \\
z_t(\mathbf{s}) &= z_{tk} \text{ if } \mathbf{s} \in P_{tk}, \\
\sigma_t^2(\mathbf{s}) &= \sigma_{tk}^2 \text{ if } \mathbf{s} \in P_{tk}, \\
\lambda &\sim N(0, \sigma_\lambda^2), \\
v_t(\mathbf{s}) \mid \Theta &\sim \text{Matérn}(\mathbf{0}, \Sigma),
\end{aligned} \tag{25}$$

where $\Theta = \{\varphi, \nu, \gamma, \lambda, \boldsymbol{\beta}\}$ and Σ is the Matérn covariance of (2). The priors on $\mathbf{w}_{tk}^*$, z_{tk}^* and σ_{tk}^{2*} follow (7)–(9) for the baseline, or (16)–(18) under our AR(2) extension. Equation (25) is the baseline of Morris et al. (2017); our contributions are the AR(2) priors above and the MRTS mean term $\mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^*$ of (24), which replace the latent dynamics and the mean while leaving the censoring layer and Matérn covariance unchanged. Predictions at new sites use Bayesian kriging on the multivariate normal full conditional of $\mathbf{Y}_t$.

3 Simulation study

In this section, we present the results from a simulation study to investigate how the AR(2) time-varying structure on latent variables and the number of thin-plate splines basis functions impact the accuracy of predictions made by the model and to compare with Gaussian and methods without the proposed extensions.

3.1 Design

All simulation designs generate data from the hierarchical model of Section 2.5 on $n_s = 144$ sites, drawn uniformly on $[0, 10] \times [0, 10]$, over $n_t = 50$ independent days. In the data generation the regression mean is held constant, $\mathbf{X}^\top \boldsymbol{\beta} = 10$ (the MRTS experiment below adds a mean-zero non-stationary random effect to one design, which leaves the regression mean at 10), while every fitted model includes an intercept and a first-order spatial trend. For each design we simulate 10 data sets, each split into 100 training sites and 44 withheld test sites, and run every MCMC chain for 20,000 iterations with a burn-in period of 10,000

iterations. Throughout, $q(0.80)$ denotes the 80th sample quantile of the data, and the five baseline analysis models are

1. Gaussian marginal, $K = 1$ knot;
2. Skew- t marginal, $K = 1$ knot, no thresholding;
3. Symmetric- t marginal, $K = 1$ knot, threshold $T = q(0.80)$;
4. Skew- t marginal, $K = 5$ knots, no thresholding;
5. Symmetric- t marginal, $K = 5$ knots, threshold $T = q(0.80)$.

Experiment for AR(2) extension. This experiment embeds a fixed AR(2) mechanism into the skew- t generating process, so the AR(2) analysis model can be tested against data with a known autoregressive structure. All four generating designs are skew- t with $\lambda = 3$, with the latent τ , $\mathbf{w}$, and z processes evolving through a fixed AR(2) recursion (16)–(18) whose coefficients are constant across time:

1. $K = 1$ knot, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
2. $K = 1$ knot, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$;
3. $K = 5$ knots, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
4. $K = 5$ knots, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$.

Both coefficient pairs lie inside the stationarity triangle (12). Each data set is fit by nine analysis models: the five baseline models above, together with four temporal variants: skew- t models with AR(1) and AR(2) time-varying coefficients (following (16)–(18)) at $K = 1$ and $K = 5$ knots.

Experiment for MRTS extension. This experiment enriches the regression component with the MRTS basis functions $f_1(\mathbf{s}), \dots, f_{K_{\text{MRTS}}}(\mathbf{s})$ of Section 2.4. We use two generating designs:

1. Skew- t , $K = 5$ knots, $\lambda = 3$, with the constant mean $\mathbf{X}^\top \boldsymbol{\beta} = 10$;
2. Skew- t , $K = 1$ knot, $\lambda = 3$, with an additive non-stationary spatial random effect $u_t(\mathbf{s})$ added on top, so that the non-stationarity lives in the second moment (covariance), not the mean.

The second design moves the non-stationarity into the dependence, to expose the structural limit of a fixed-effect mean basis. It adds a mean-zero, per-day random effect built from the cosine basis of Tzeng and Huang (2018), written natively on the $[0, 10]^2$ domain (no rescaling to the unit square):

- Two fixed cosine fields with centres $\mathbf{c}_1 = (0, 10)^\top$, $\mathbf{c}_2 = (7.5, 2.5)^\top$ and native wavenumbers $\kappa_1 = \pi/10$, $\kappa_2 = \pi/5$,

$$\phi_j(\mathbf{s}) = \cos(\kappa_j \|\mathbf{s} - \mathbf{c}_j\|), \quad j = 1, 2. \quad (26)$$

(With $\kappa_j = a_j\pi/10$ and $\mathbf{c}_j$ on $[0, 10]^2$, $a_1 = 1$, $a_2 = 2$, this is the native form of the cosine bumps of Tzeng and Huang (2018), avoiding any rescaling to the unit square.)

- The random effect superposes them with independent, daily redrawn Gaussian weights,

$$u_t(\mathbf{s}) = \sum_{j=1}^2 \xi_{tj} \phi_j(\mathbf{s}), \quad \xi_{tj} \stackrel{iid}{\sim} N(0, \tau_j^2), \quad (\tau_1, \tau_2) = (5, 3), \quad (27)$$

independently across j and across days t .

- The response adds u_t to the 1-knot skew- t model,

$$Y_t(\mathbf{s}) = 10 + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s}) + u_t(\mathbf{s}). \quad (28)$$

- Marginalising over ξ , u_t has zero mean and the low-rank covariance $\text{Cov}(u_t(\mathbf{s}), u_t(\mathbf{s}')) = \sum_j \tau_j^2 \phi_j(\mathbf{s}) \phi_j(\mathbf{s}')$, whose variance $\sum_j \tau_j^2 \phi_j(\mathbf{s})^2$ varies with location: the dependence is non-stationary while the mean stays constant at 10.
- Because u_t is redrawn each day and has zero time-average, no fixed mean function exists for the time-pooled β^* of (24) to recover. This design is a deliberate negative control: the MRTS augmentation acts on the first moment, so it cannot represent a non-stationary second-moment structure, and the score- and recovery-versus- K curves should stay flat or rise.

We emphasise the honest scope this sets up. The MRTS basis is a fixed-effect mean (trend) component; it reduces mean estimation error only when the structure genuinely lies in the mean, and any sufficiently rich basis would do the same—the MRTS-specific benefit is the tuning-free, coarse-to-fine ordering of K_{MRTS} , not a unique gain in mean accuracy. The

resolution-adaptive property for which MRTS was designed by Tzeng and Huang (2018) applies to low-rank covariance (fixed-rank kriging) with random coefficients, which the fixed-effect mean use here does not engage. The random-effect design makes this concrete: when the non-stationarity is a second-moment (covariance) feature, the mean augmentation has no purchase on it.

Each of the five baseline models is fit across MRTS basis sizes $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$, where $K_{\text{MRTS}} = 0$ recovers the no-MRTS baseline.

3.2 Cross validation

Models are compared using cross validation, with 100 sites used as training sites to fit each setting and 44 sites withheld for evaluating predictions on each replicate. Because one of the primary goals of this model is to predict exceedances over a high level, we use Brier scores to compare the models (Gneiting and Raftery, 2007). For an exceedance level L , the Brier score is

$$\text{BS}(L) = \{ I(y > L) - \hat{P}(Y > L) \}^2, \quad (29)$$

where y is the held-out observation, $I(y > L)$ is the exceedance indicator, and $\hat{P}(Y > L)$ is the posterior predictive probability of exceeding L . We average the Brier scores over all test sites and days; lower values indicate better exceedance prediction.

3.3 Results on AR(2) comparison

In general, we find that the methods whose marginal matches the generating truth give the largest improvement over the Gaussian model, while a mismatched model can perform worse than Gaussian, particularly in the far tail (Figure 1). The preferred knot count tracks the data-generating truth rather than the strength of the temporal dependence: the one-knot models are best when the data are generated with one knot, and the five-knot models are best when the data are generated with five knots, with the AR(2) coefficients shifting the magnitude of the scores but not their ordering.

Adding AR(2) (or AR(1)) temporal structure does not help. Even on data explicitly generated with an AR(2) recursion (the most favorable possible setting for the temporal extension), the temporal curves sit essentially on top of their non-temporal counterparts at every quantile and in every setting. Predictive accuracy is governed by the flexibility of the marginal and

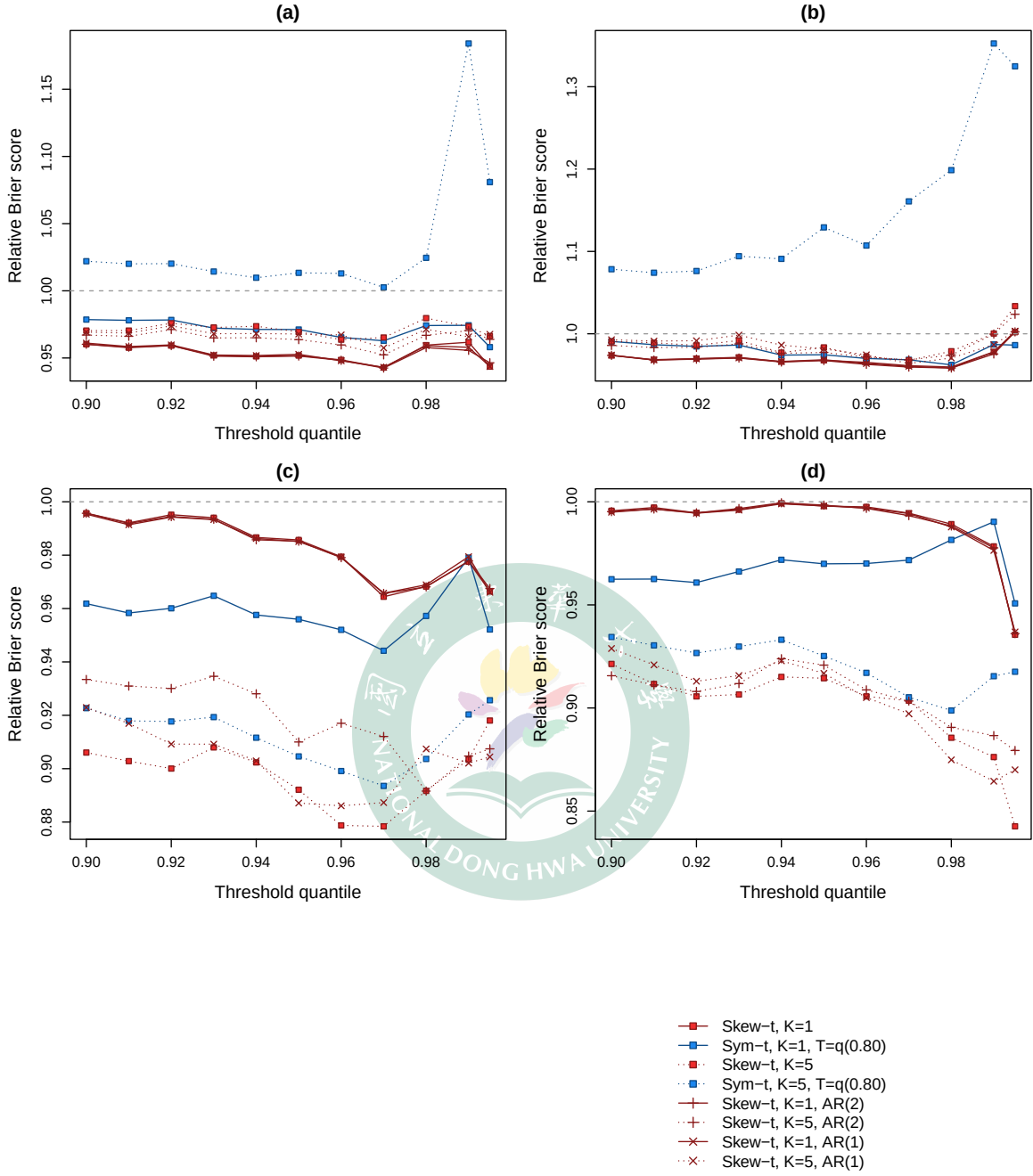


Figure 1: Relative Brier score versus the Gaussian model by threshold quantile (Gaussian is the reference $\equiv 1$, shown only as the dashed line; values below it are better). Data-generating settings: (a) skew- t ($K=1$, $\lambda=3$), AR(2): $\phi_1=0.8$, $\phi_2=-0.35$; (b) skew- t ($K=1$, $\lambda=3$), AR(2): $\phi_1=0.12$, $\phi_2=-0.05$; (c) skew- t ($K=5$, $\lambda=3$), AR(2): $\phi_1=0.8$, $\phi_2=-0.35$; (d) skew- t ($K=5$, $\lambda=3$), AR(2): $\phi_1=0.12$, $\phi_2=-0.05$.

the spatial partition, namely the skew- t margin and the number of knots, rather than by the autoregressive order of the latent coefficients.

The recovery of the AR(2) coefficients tells the same story (Table 1). Only the scale process is recovered with any fidelity, and only under strong dependence at a matched one-knot fit; the skewness process is recovered only partially and the knot-location process is essentially noise. Recovery deteriorates as the knot count grows and as the dependence weakens, so even on AR(2)-generated data the temporal signal is only weakly identified. In summary, the AR(2) structure should be treated as a candidate option rather than a default improvement.

Table 1: Posterior mean (SD) of the recovered AR(2) coefficients in the four fixed-AR(2) settings. Gen K is the data-generating knot count and Fit K the analysis-model knot count; $\hat{\phi}^{(\sigma)}$, $\hat{\phi}^{(z)}$, $\hat{\phi}^{(w)}$ are the estimated AR(2) coefficients of the scale, skewness, and knot-location processes ((16)–(18)).

Settings		Fit K	$\hat{\phi}^{(\sigma)}$		$\hat{\phi}^{(z)}$		$\hat{\phi}^{(w)}$	
(ϕ_1, ϕ_2)	K		$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$
(0.80, −0.35)	1	1	0.803(0.140)	−0.363(0.137)	0.626(0.114)	−0.225(0.157)	-	-
(0.80, −0.35)	1	5	0.871(0.142)	−0.438(0.164)	0.472(0.284)	−0.270(0.162)	−0.129(0.377)	0.062(0.257)
(0.12, −0.05)	1	1	0.083(0.122)	−0.019(0.150)	0.186(0.137)	0.084(0.226)	-	-
(0.12, −0.05)	1	5	0.095(0.141)	−0.050(0.220)	0.050(0.088)	0.051(0.142)	0.002(0.215)	0.179(0.259)
(0.80, −0.35)	5	1	0.512(0.190)	0.043(0.245)	0.434(0.185)	−0.108(0.128)	-	-
(0.80, −0.35)	5	5	0.169(0.115)	−0.023(0.118)	0.074(0.093)	−0.006(0.090)	−0.019(0.206)	0.094(0.135)
(0.12, −0.05)	5	1	0.285(0.228)	0.161(0.164)	0.163(0.122)	−0.063(0.122)	-	-
(0.12, −0.05)	5	5	−0.001(0.072)	−0.017(0.137)	0.010(0.075)	−0.010(0.067)	−0.017(0.193)	0.115(0.190)

3.4 Results on MRTS comparison

In general, augmenting the regression component with MRTS basis functions does not deliver a stable improvement in upper-tail scores (Figure 2). As with the AR(2) comparison, the data-generating knot count dominates the ranking: the matching five-knot skew- t model is best on the five-knot setting, and the one-knot model is best on the random-effect process.

Enlarging the MRTS basis does not lower the mean Brier score on either setting. On the five-knot setting every curve is essentially flat in K_{MRTS} . On the additive non-stationary spatial random-effect process the curves are likewise flat, with over-specified models destabilising at large K_{MRTS} . This second setting is the decisive one: the non-stationarity lives in the second moment (a daily-redrawn random effect u_t), whereas the MRTS basis enters only the fixed-effect mean. Because u_t has zero time-average, the time-pooled coefficient

β^* has no first-moment structure to recover, so the Brier- and recovery-versus- K curves stay flat by construction—not for want of signal, since u_t contributes substantial (spatially non-stationary) variance. The MRTS mean term therefore yields no reliable tail-score gain: a fixed-effect mean cannot represent a non-stationary second-moment feature, and so the spatial mean is not the binding constraint on upper-tail prediction.

Overall, the simulation study indicates that both extensions enlarge the usable model class but that neither uniformly dominates the STP baseline. The AR(2) prior provides temporal flexibility and the MRTS covariates enrich the spatial regression component, yet their empirical benefits depend on the data-generating mechanism and on the target exceedance level. Whether the same pattern holds on real environmental data (and, if so, what it implies for practitioners choosing among the candidate settings) is the question taken up in the Data analysis section, where AR(2) and MRTS variants are treated as candidate options to be selected by held-out predictive scores rather than as default replacements for the AR(1) STP baseline.

4 Real data analysis

Morris et al. (2017) introduced the spatio-temporal skew- t (STP) process to predict high-threshold ground-level ozone exceedances from U.S. EPA monitoring data. We return to that same data here, asking not for a new model class but whether enlarging the STP candidate class along the two axes developed in this thesis, an AR(2) temporal prior and an MRTS spatial-mean term, yields better held-out tail prediction. The comparison is therefore designed to identify the exceedance levels, if any, at which the enlarged candidate class improves on the STP baseline, rather than to select a single overall-best model.

The response is the daily maximum 8-hour ozone concentration for the 31 days of July 2005 at United States Air Quality System (AQS) monitoring sites, with the co-located Community Multi-scale Air Quality (CMAQ) model output as the sole covariate. Following Morris et al. (2017), the regression component is $\mathbf{X}_t(\mathbf{s}) = [1, \text{CMAQ}_t(\mathbf{s})]^\top$, so that each model acts as a statistical calibration of CMAQ against the monitor observations.

4.1 Model comparisons

The baseline candidate set follows that of Morris et al. (2017). We fit Gaussian and skew- t marginals over a range of partition sizes, $K \in \{1, 5, 6, 7, 8, 9, 10, 15\}$, and censor the response at three levels to contrast no, moderate, and high censoring: no censoring ($T = 0$), a moderate threshold ($T = 50$ ppb, the 0.42 sample quantile), and a high threshold ($T = 75$ ppb, the 0.92 sample

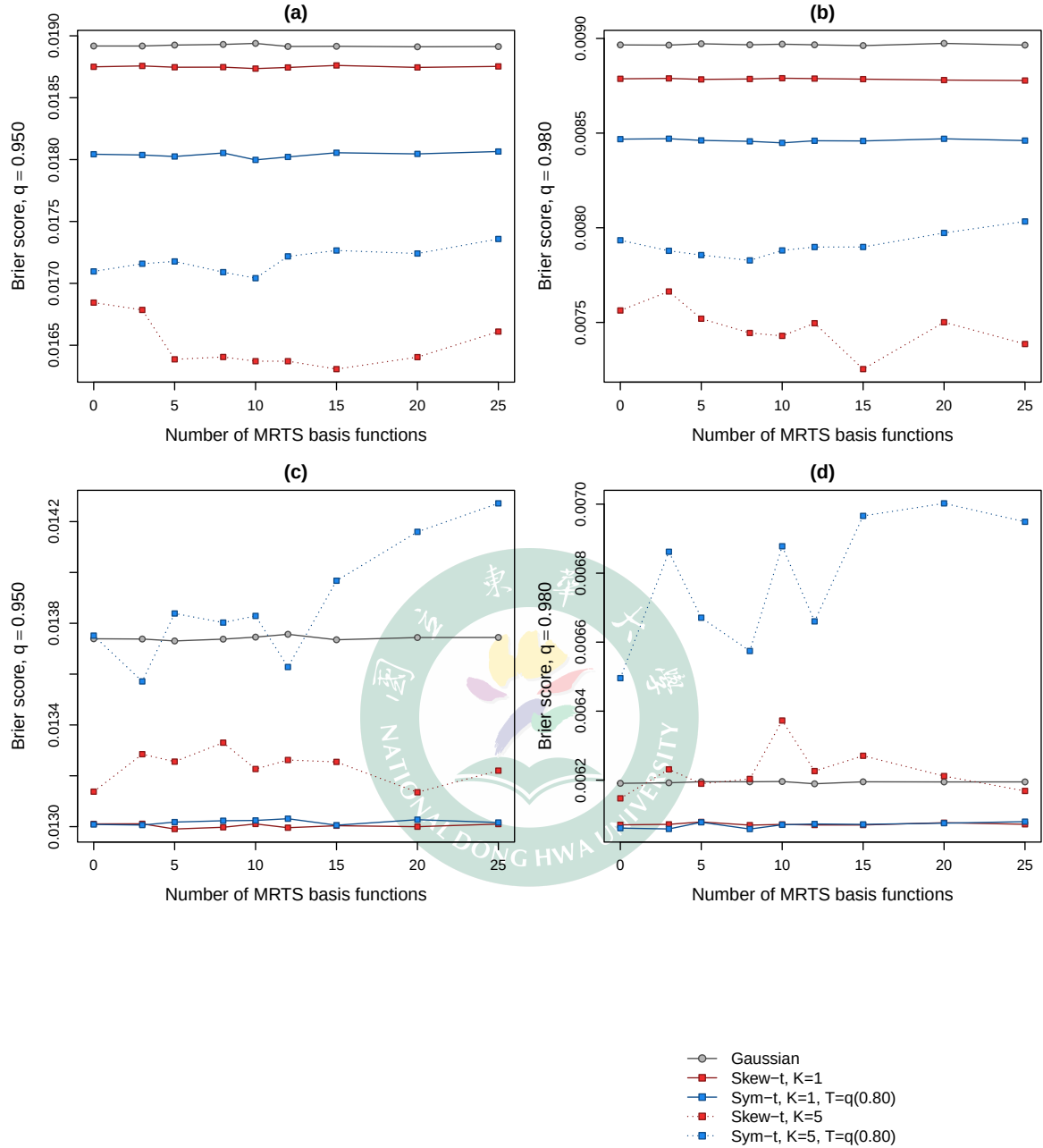


Figure 2: Brier score versus the number of MRTS basis functions K_{MRTS} ($K_{\text{MRTS}} = 0$ is the no-MRTS baseline; lower is better). Data-generating processes: (a,b) skew- t ($K = 5$, $\lambda = 3$); (c,d) skew- t ($K = 1$, $\lambda = 3$) with an additive non-stationary spatial random effect. Exceedance level $q = 0.95$ in (a,c) and $q = 0.98$ in (b,d).

quantile). The 75 ppb level is of regulatory interest because the United States air-quality standard is defined by exceedance of 75 ppb; for settings censored at $T = 75$ we use a symmetric- t marginal, as in the simulation study. Each setting is fit both with and without a latent time series.

On top of this baseline we add the two extensions developed in this thesis. The first replaces the AR(1) latent dynamics with the standardized AR(2) prior of Section 2.3; the second augments the regression component with MRTS basis functions, so that non-stationary spatial mean structure can be absorbed by multiple-resolution thin-plate splines.

Each MCMC chain is run for 30,000 iterations with a burn-in period of 25,000 iterations. The comparison draws on 800 monitoring sites and is evaluated by two-fold cross validation, with 400 training and 400 validation sites per fold, scored by the held-out Brier score of Section 3.2.

4.2 Results

Across the upper tail, the proposed extensions surface among the best models at scattered exceedance levels rather than improving prediction uniformly, so no single extension, and no single setting, is best across the whole tail (Table 2). The extensions nonetheless earn their place by supplying the leading model at specific points along the tail that the baseline alone does not reach.

Table 2: Top two performing models for predicting ozone exceedance of level L with Brier score.

Level L	Rank 1 model	Rank 2 model
0.900	MRTS skew- t , $K = 1$, $T = 0$, AR(1), $K_{\text{MRTS}} = 10$	skew- t , $K = 1$, $T = 0$, AR(1)
0.950	AR2 skew- t , $K = 7$, $T = 50$	AR2 skew- t , $K = 10$, $T = 50$
0.980	skew- t , $K = 6$, $T = 50$, AR(1)	skew- t , $K = 5$, $T = 50$, AR(1)
0.990	symmetric- t , $K = 9$, $T = 75$, AR(1)	skew- t , $K = 5$, $T = 50$
0.995	AR2 symmetric- t , $K = 15$, $T = 75$	symmetric- t , $K = 9$, $T = 75$, AR(1)

This picture is reproducible on smaller training samples. Repeating the comparison under two further cross-validation splits, written $N_{\text{train}}/N_{\text{test}}$ for N_{train} training and N_{test} validation sites (the 300/100 and 200/200 splits; Tables 3 and 4), recovers the same qualitative ranking: the Baseline, AR(2), and MRTS families continue to share the top ranks across the tail, even as the exact best-ranked setting shifts with the split and the level. That shifting is expected, since only a handful of validation sites exceed the highest thresholds and the fine ranking

Table 3: Top four Brier score models for the 300/100 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions.

Level L	Rank 1	Rank 2	Rank 3	Rank 4
0.900	MRTS (0.041)	Baseline (0.041)	MRTS (0.041)	Baseline (0.041)
0.950	Baseline (0.020)	Baseline (0.020)	Baseline (0.020)	Baseline (0.020)
0.980	MRTS (0.008)	AR2 (0.008)	Baseline (0.008)	AR2 (0.008)
0.990	AR2 (0.003)	Baseline (0.003)	Baseline (0.003)	Baseline (0.003)
0.995	Baseline (0.001)	Baseline (0.001)	Baseline (0.001)	AR2 (0.001)



Table 4: Top four Brier score models for the 200/200 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions.

Level L	Rank 1	Rank 2	Rank 3	Rank 4
0.900	Baseline (0.050)	AR2 (0.050)	Baseline (0.050)	MRTS (0.050)
0.950	AR2 (0.030)	AR2 (0.030)	AR2 (0.030)	Baseline (0.030)
0.980	MRTS (0.015)	MRTS (0.015)	AR2 (0.015)	Baseline (0.015)
0.990	Baseline (0.008)	AR2 (0.008)	AR2 (0.008)	Baseline (0.008)
0.995	Baseline (0.004)	Baseline (0.004)	Baseline (0.004)	AR2 (0.004)

there is sensitive to the held-out draw; the persistence of the broad pattern across splits is what supports retaining AR(2) and MRTS as candidates.

5 Discussion

This thesis asked not whether the spatio-temporal skew- t process predicts high-threshold exceedances well, which Morris et al. (2017) already established, but where within its four components room for sharper upper-tail prediction remains. To probe it, we enlarged the model along two opposing axes: an AR(2) latent prior adding temporal memory, and an MRTS mean term adding a flexible non-stationary mean (a spatial trend). We scored both on held-out data rather than asserting either as a default. The temporal axis yielded almost no usable signal: even on AR(2)-generated data the temporal curves sat on top of their non-temporal counterparts at every quantile, and the recovered coefficients (Table 1) were only weakly identified, because tail prediction is set by how faithfully the marginal and the spatial partition capture the upper tail, not by the memory of the latent coefficients. The spatial axis was more favorable but no more stable, with the MRTS basis improving selected scores yet failing to hold across families or quantile levels; and while the far-tail ranking is sensitive to the handful of sites exceeding the highest thresholds, so that the single best setting shifts with the split, the broad pattern in which the baseline, AR(2), and MRTS families recur among the top ranks persists across all splits, and it is that persistence that justifies retaining the extensions.

The lesson reaches past ozone, since the same target, a spatial probability statement about the upper tail at unmonitored sites, arises for any heavy-tailed, asymmetric environmental field regulated by a high threshold, from precipitation extremes to heat warnings: temporal order and mean flexibility are not free improvements to such a model, but neither are they idle. Rather than replacing the STP baseline, the standardized AR(2) prior and the MRTS mean enlarge it into a candidate class from which held-out tail scores can select the setting that fits the data at hand, giving practitioners who must predict high-threshold exceedances a broader, data-driven choice than any single fixed model can offer.

A Appendices

A.1 Stationary moments of the standardized AR(2)

We derive the Yule–Walker moments (13)–(14) used to standardize the AR(2) prior (11) to unit marginal variance. Let $\gamma_h = \text{Cov}(Y_t, Y_{t-h})$ be the stationary autocovariances, normalized so that $\gamma_0 = \text{Var}(Y_t) = 1$. Multiplying the recursion (11) by Y_{t-h} for $h \geq 1$ and taking expectations, and using that ϵ_t is independent of Y_{t-h} , yields the Yule–Walker recursion

$$\gamma_h = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}, \quad h \geq 1. \quad (30)$$

Evaluating (30) at $h = 1$ and using the symmetry $\gamma_{-1} = \gamma_1$ with $\gamma_0 = 1$ gives $\gamma_1 = \phi_1 + \phi_2 \gamma_1$, hence $\gamma_1 = \phi_1 / (1 - \phi_2)$, and at $h = 2$ gives $\gamma_2 = \phi_1 \gamma_1 + \phi_2$; together these are (13). Multiplying (11) instead by Y_t and taking expectations, with $E[Y_t \epsilon_t] = \sigma_\epsilon^2$ because Y_t carries the current innovation, gives $\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma_\epsilon^2$, and solving for σ_ϵ^2 under $\gamma_0 = 1$ recovers (14). The same normalization fixes the initial pair: (Y_1, Y_2) has unit variances and lag-1 covariance γ_1 , which is the stationary bivariate Gaussian (15).

A.2 AR(2) full conditional for the Gibbs-within-MH update

The AR(2) Markov structure factorizes the joint prior of $(Y_1, \dots, Y_{n_t})$ as

$$p(Y_1, \dots, Y_{n_t}) = p(Y_1, Y_2) \prod_{s=3}^{n_t} p(Y_s | Y_{s-1}, Y_{s-2}), \quad (31)$$

with $p(Y_1, Y_2)$ given by (15) and each conditional factor for $s \geq 3$ following (11). For an interior time point $3 \leq t \leq n_t - 2$, the value Y_t enters (31) through three factors: the self factor $p(Y_t | Y_{t-1}, Y_{t-2})$, the lag-1 factor $p(Y_{t+1} | Y_t, Y_{t-1})$, and the lag-2 factor $p(Y_{t+2} | Y_{t+1}, Y_t)$. The full conditional in the Gibbs-within-MH update of Y_t is therefore

$$\pi(Y_t | Y_{-t}, \mathbf{Y}) \propto L(\mathbf{Y} | Y_t) \cdot \underbrace{p(Y_t | Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} | Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} | Y_{t+1}, Y_t)}_{\text{lag-2}}, \quad (32)$$

where $L(\mathbf{Y} | Y_t)$ is the spatial likelihood contributed by Y_t through the partition and Gaussian process structure of (1). The lag-2 factor is absent for $t = n_t - 1$ and both lag-1 and lag-2 factors are absent for $t = n_t$. Under a symmetric Gaussian random-walk proposal $Y_t^{(c)} \sim$

$N(Y_t, s^2)$, the Metropolis–Hastings acceptance ratio is

$$R = \min\left(1, \frac{\pi(Y_t^{(c)} | Y_{-t}, \mathbf{Y})}{\pi(Y_t | Y_{-t}, \mathbf{Y})}\right), \quad (33)$$

in which all three prior factors must appear; omitting the lag-2 factor distorts the stationary distribution of the chain away from the AR(2) joint and produces a lag-2 autocorrelation that does not match the Yule–Walker value in (13).

References

- Coles, Stuart, Janet Heffernan, and Jonathan Tawn (1999). “Dependence Measures for Extreme Value Analyses”. In: *Extremes* 2.4, pp. 339–365. ISSN: 1572-915X. DOI: 10.1023/A:1009963131610. URL: <https://doi.org/10.1023/A:1009963131610>.
- Gneiting, Tilmann and Adrian E Raftery (Mar. 2007). “Strictly Proper Scoring Rules, Prediction, and Estimation”. In: *Journal of the American Statistical Association* 102.477. _eprint: <https://doi.org/10.1198/016214506000001437>, pp. 359–378. ISSN: 0162-1459. DOI: 10.1198/016214506000001437. URL: <https://doi.org/10.1198/016214506000001437> (visited on 11/10/2025).
- Morris, Samuel A. et al. (Sept. 2017). “A Space-Time Skew-t Model for Threshold Exceedances”. In: *Biometrics* 73.3, pp. 749–758. ISSN: 0006-341X. DOI: 10.1111/biom.12644. URL: <https://doi.org/10.1111/biom.12644> (visited on 03/14/2025).
- Tzeng, ShengLi and Hsin-Cheng Huang (Apr. 2018). “Resolution Adaptive Fixed Rank Kriging”. In: *Technometrics* 60.2. _eprint: <https://doi.org/10.1080/00401706.2017.1345701>, pp. 198–208. ISSN: 0040-1706. DOI: 10.1080/00401706.2017.1345701. URL: <https://doi.org/10.1080/00401706.2017.1345701> (visited on 03/22/2026).
- U.S. Environmental Protection Agency (Oct. 2024). *Bayesian Space-time Downscaling Fusion Model (Downscaler) Derived Estimates of Air Quality for 2021*. Annual Report EPA-454/R-24-002. Office of Air Quality Planning and Standards, U.S. EPA. URL: <https://www.epa.gov/hesc/rsig-related-downloadable-data-files>.